

DevOps Monitoring Sandbox – Documentation

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1. Introduction

This project sets up a complete **monitoring and alerting sandbox environment** using Docker containers.

The stack includes:

- **Prometheus** → Metrics collection
- **Node Exporter** → System metrics exporter
- **Grafana** → Dashboard visualization
- **Alertmanager** → Alerts handling

2. Steps Performed (Complete Workflow)

2.1 Installed and Started Docker Desktop

- Downloaded Docker Desktop on macOS (Apple Silicon).
- Started Docker Desktop (ensured the whale icon shows *Running*).
- Verified Docker Engine running:

```
docker info
```

2.2 Created DevOps Sandbox Project Folder

```
mkdir devops-sandbox  
cd devops-sandbox
```

2.3 Created Docker Compose File

Created `docker-compose.yml` with the following services:

- Prometheus (port **9090**)
- Grafana (port **3000**)

- Alertmanager (port **9093**)
- Node Exporter (port **9100**)

docker-compose.yml

```

services:
  prometheus:
    image: prom/prometheus:latest
    ports:
      - "9090:9090"
    volumes:
      - ./prometheus.yml:/etc/prometheus/prometheus.yml

  grafana:
    image: grafana/grafana:latest
    ports:
      - "3000:3000"
    volumes:
      - grafana-storage:/var/lib/grafana

  alertmanager:
    image: prom/alertmanager:latest
    ports:
      - "9093:9093"
    volumes:
      - ./alertmanager.yml:/etc/alertmanager/alertmanager.yml

  node-exporter:
    image: prom/node-exporter:latest
    ports:
      - "9100:9100"

volumes:
  grafana-storage:

```

2.4 Created Prometheus Configuration File

Created file:

```
nano prometheus.yml
```

Added:

```

global:
  scrape_interval: 5s

```

```
scrape_configs:
  - job_name: "prometheus"
    static_configs:
      - targets: ["localhost:9090"]

  - job_name: "node_exporter"
    static_configs:
      - targets: ["node-exporter:9100"]
```

2.5 Created Alertmanager Configuration File

```
nano alertmanager.yml
Added:
```

```
route:
  receiver: default

receivers:
  - name: default
    webhook_configs:
      - url: http://localhost:5001/
```

2.6 Started the Monitoring Stack

Ran:

```
docker compose down
docker compose up -d
All services successfully started.
```

Verified with:docker ps

Got:

- 9090->9090 prometheus 3000->3000 grafana 9093->9093 alertmanager 9100->9100 node-exporter

2.7 Confirmed Prometheus Targets Working

Checked targets:

```
curl http://localhost:9090/api/v1/targets
```

Output showed:

- **node-exporter:9100 → UP**
- **localhost:9090 → UP**

Metrics confirmed:node_cpu_seconds_total{...}

2.8 Configured Grafana

Opened: <http://localhost:3000>

Login:admin / admin

Added Prometheus as a data source:

URL:http://prometheus:9090

Clicked **Save & Test** → **✓ Data source is working**

2.9 Imported Official Grafana Dashboard

Dashboard ID: **1860**

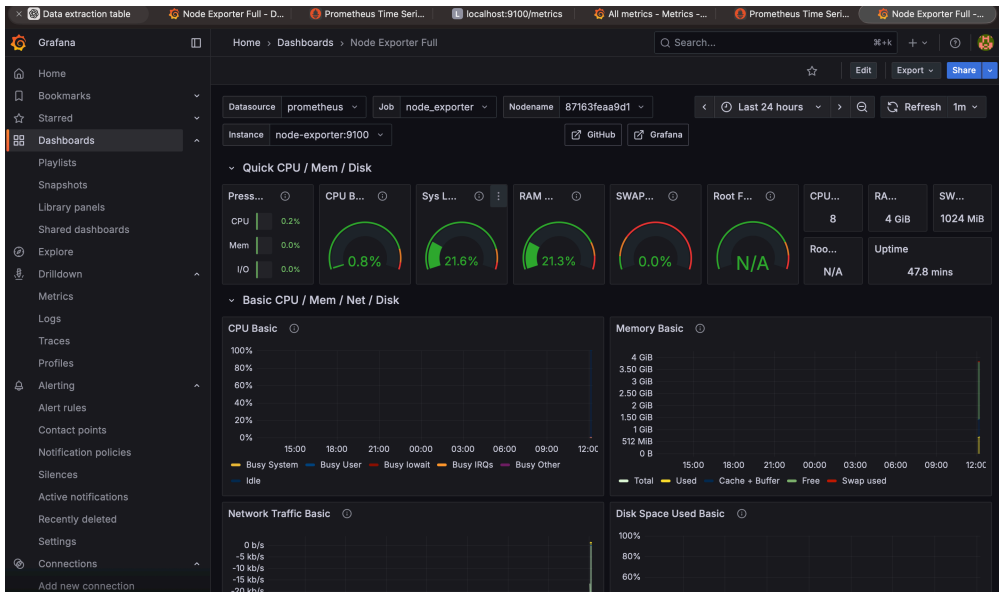
Steps:

- Grafana → Dashboards → Import
- Enter 1860
- Select Prometheus data source
- Imported "Node Exporter Full Dashboard"

Now CPU, memory, disk, network graphs display real-time data.

3. Final Working System

Services running:



Name	port	Purpose
Grafana	3000	Visualization dashboards
Prometheus	9090	Metrics scraping
Alertmanager	9093	Alerts
Node Exporter	9100	System metrics

Parameters scraped:

- CPU usage
- Memory usage
- Disk usage
- Network I/O
- Load average
- Filesystem metrics
- Prometheus internal metrics

4. Conclusion

I have successfully built a complete DevOps Monitoring Sandbox using Docker:

- 1.Containers running
- 2.Prometheus scraping data
- 3.Grafana dashboards working
- 4.Node Exporter metrics visible
- 5.Alertmanager configured

This sandbox is ready for learning, demos, and automation practices.

```
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % rm alert.manager.yml.save
rm: alert.manager.yml.save: No such file or directory
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % rm alertmanager.yml.save
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % ls
alertmanager.yml    configs    dashboards    docker-compose.yml    prometheus.yml    provision
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % docker compose up -d
[+] Running 5/5
   Network devops-sandbox_default                  Created                                0.0s
   Container devops-sandbox-node-exporter-1         Started                                0.3s
   Container devops-sandbox-alertmanager-1          Started                                0.3s
   Container devops-sandbox-grafana-1               Started                                0.3s
   Container devops-sandbox-prometheus-1            Started                                0.3s
Error response from daemon: failed to create task for container: failed to create shim task: OCI runtime create failed: runc create failed: unable to start container process: error during container init
: error mounting "/host_mnt/Users/devendrakumar/devops-sandbox/prometheus.yml" to rootfs at "/etc/prometheus/prometheus.yml": create mountpoint for /etc/prometheus/prometheus.yml mount: cannot create su
bdirectories in /var/lib/docker/containers/overlays/8401da64d18c7f63d5e728dace2ab464b4b27f586ad285a7fc31c2941fbd96/etc/prometheus/prometheus.yml: not a directory: unknown: Are you trying to mount a dir
ectory onto a file (or vice-versa)? Check if the specified host path exists and is the expected type
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % ls
alertmanager.yml    configs    dashboards    docker-compose.yml    prometheus.yml    provision
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
7f6bba3f2c8    grafana/grafana:latest              /run.sh                 25 seconds ago Up 25 seconds    0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp    devops-sandbox-grafana-1
430b3d47d61    prom/node-exporter:latest           /bin/node_exporter      25 seconds ago Up 25 seconds    0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp    devops-sandbox-node-exporter-1
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % docker compose down
[+] Running 5/5
   Container devops-sandbox-node-exporter-1         Removed                                0.1s
   Container devops-sandbox-grafana-1               Removed                                0.2s
   Container devops-sandbox-prometheus-1            Removed                                0.0s
   Container devops-sandbox-alertmanager-1          Removed                                0.0s
   Network devops-sandbox_default                   Removed                                0.1s
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % ls
alertmanager.yml    configs    dashboards    docker-compose.yml    prometheus.yml    provision
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % rm -r alertmanager.yml
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % rm -r prometheus.yml
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % ls
configs    dashboards    docker-compose.yml    provision
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % nano prometheus.yml
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % nano alertmanager.yml
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % ls
alertmanager.yml    configs    dashboards    docker-compose.yml    prometheus.yml    provision
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % docker compose up -d
[+] Running 5/5
   Network devops-sandbox_default                  Created                                0.0s
   Container devops-sandbox-alertmanager-1         Started                                0.3s
   Container devops-sandbox-node-exporter-1         Started                                0.3s
   Container devops-sandbox-prometheus-1            Started                                0.3s
   Container devops-sandbox-grafana-1               Started                                0.3s
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
85673ac6eb0e    prom/prometheus:latest              /bin/prometheus --c...  8 seconds ago Up 7 seconds    0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp    devops-sandbox-prometheus-1
7238d6d92d8f    grafana/grafana:latest              /run.sh                 8 seconds ago Up 7 seconds    0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp    devops-sandbox-grafana-1
effb95a3880    prom/alertmanager:latest            /bin/alertmanager --...  8 seconds ago Up 7 seconds    0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp    devops-sandbox-alertmanager-1
87163faa9d1    prom/node-exporter:latest           /bin/node_exporter      8 seconds ago Up 7 seconds    0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp    devops-sandbox-node-exporter-1
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % curl http://localhost:9090/api/v1/targets
{"status":"success","data":{"activeTargets":[{"discoveredLabels":{"__address__":"node-exporter:9100","__node-exporter:9100","__metrics_path__":"/metrics","__scheme__":"http","__scrape_interval__":"5s","__scrape_timeout__":"5s","__job__":"node_exporter"},"labels":{"instance":"node-exporter:9100"},"lastScrape":{"@version":1,"lastScrapeDuration":0.0367215,"health":"up","scrapeInterval":"5s","scrapeTimeout":"5s"},"discoveredLabels":{"__address__":"loc
alhost:9090","__metrics_path__":"/metrics","__scheme__":"http","__scrape_interval__":"5s","__scrape_timeout__":"5s"},"job":"prometheus"},"labels":{"instance":"localhost:9090"},"scrape":
{"@version":1,"scrapeInterval":"5s","scrapeTimeout":"5s"},"droppedTargetCounts":{"node_exporter":0,"prometheus":0}}]}
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox % ls
alertmanager.yml    configs    dashboards    docker-compose.yml    prometheus.yml    provision
devendrakumar@DEVENDRAS-MacBook-Air devops-sandbox %
```